Prathamesh Mandke

+1(540) 252 9660 • pkmandke@vt.edu • https://pkmandke.github.io
pkmandke • San Diego, CA

Education

Virginia Polytechnic Institute and State University **Blacksburg, VA**
Master of Engineering, Computer Engineering, GPA: 4.0/4.0 *Aug 2019 – May 2021*
Relevant Coursework:

- Deep Learning
- Computer Vision
- Computation for Data Science
- Advanced Machine Learning
- Advanced Parallel Computing
- Information Storage & Retrieval

College of Engineering, Pune **Pune, India**
B.Tech Electronics & Telecommunication GPA: 9.11/10 *Aug 2015 – May 2019*
Minor in Computer Engineering

Experience

Qualcomm, Inc. **San Diego, CA**
Machine Learning Engineer *Jul 2021 – Present*

- **Keywords**: Embedded AI, Machine Learning, Software Development, Federated Learning (FL), On-Device Learning
- Working with the Federated and On-Device Learning team at Qualcomm AI Research
- Significant contributions to QCOMs Snapdragon FL framework including gRPC modules, monitoring dashboards, etc.
- Key Developer for Federated User Verification (FedUV) demo using 25 Android devices showcased at NeurIPS 2021. Source: https://www.youtube.com/watch?v=fTVmDzFKjqI
- Co-authored US Patent granted for Federated Learning metrics aggregation logic
- Designed and developed initial version of the On-Device Learning library from scratch in C++/Python
- Key Developer working with our external collaborator on FL FedUV. Work accepted at InterSpeech'23 conference.
- **Skills/Tools**: gRPC, Python, C++, Kotlin

Virginia Tech **Blacksburg, VA**
Graduate Research Assistant *Sept 2020 – May 2021*

- **Domain**: Deep Learning for Scientific Applications
- Worked with Prof. Anuj Karpatne in the SGML Lab at the Dept of Computer Science
- Part of the NSF funded HDR-BGNN project on augmenting Machine Learning models with scientific data in form of phylogenic trees and ontologies
- Worked on the problem of weakly supervised semantic segmentation using domain specific priors.
- **Keywords**: Deep Learning, Semantic Segmentation, ML Research

Qualcomm, Inc. **(Remote) San Diego, CA**
ML Software Engineer Intern *Summer 2020*

- **Keywords**: Deep Learning, Software Development, Model Quantization
- Worked on the AI Model Efficiency Toolkit (AIMET) team: https://github.com/quic/aimet
- Developed visualization utilities for Quantization Simulation & Data Free Quantization, & integrated with Netron.
- Implemented a generic utility to convert AIMET's internal model graph representation to ONNX.

Flytbase, Inc. **Pune, India**
Internship *Summer 2019*

- **Domain**: Deep Learning, Autonomous Drones, Software Development
- Worked on barcode localization for warehouse automation applications.
- Curated a custom barcode dataset using an in-house DGI drone for ML training.
- Trained Yolo, Faster RCNN and SSD models with Inception, ResNet and MobileNet backbones.
- Explored embedded deployment of models on the Intel Neural Compute Stick using docker in Linux.
- **Keywords**: Deep Learning, Autonomous Systems, Object Detection

Siemens, Ltd. **Mumbai, India**
Internship *Summer'17, Summer'18*

- **Domain**: Industrial Automation
- Re-vamped design, power circuit & completed programming of Programmable Logic Controller (PLCs) for contactor testing automatons to achieve significant cycle time reduction.
- **Keywords**: Programmable Logic Controller, Ladder logic

Projects

Deep Knowledge Transfer: CNN Model Compression for OpenCL-FPGA deployment **Dec'18 - May'19**

- Bachelor's Thesis: Explored knowledge distillation in the regression based FaceNet CNN for model compression.
- Inception based teacher networks used to train 75-58% smaller MobileNet student models on the ~1M VGG2 face dataset using knowledge distillation to achieve 80-83% accuracies on the student models
- OpenCL Kernels for each layer type in the teacher (Inception) and student (MobileNet) models deployed on Intel's DE10 Nano FPGA SoC for CNN inference.
- **Skills**: Python, Tensorflow, OpenCV, OpenCL. Details: [github].

Image Reconstruction in Shape Controlled Cell Migration using Deep Learning **Jan'21 - May'21**

- Master's Project: Image reconstruction of cell migration imagery in fibrous environments.
- Worked on the problem of recovering fluorescent images of cell structures from phase images using Deep Learning.
- Used convolutional auto-encoders, pix-2-pix and CycleGAN architectures for baselines.
- Employed advanced attention based GAN methods to improve performance over baselines.
- **Skills**: Python, PyTorch, Linux, GPU based training.

Sparse View CT Image Reconstruction using Deep CNNs **Jan'20 – May'20**

- Modified U-Net CNN architecture with transpose convolutions and residual connection to achieve robust reconstruction performance in terms of SSIM and PSNR.
- Implemented Wasserstein GAN based reconstruction pipelines in PyTorch and evaluated against vanilla GANs.
- Poster accepted to AAPM — COMP 2020 [URL]

Human Posture Recognition using Artificial Neural Networks **Feb'18 - May'18**

- A system to classify human postures on a Raspberry-Pi using an Artificial Neural Network.
- Designed & built PCB node to interface ESP8266 w/ MPU-6050 IMU sensor to transmit data to a Raspberry-pi.
- Used 2 sensor nodes(thigh and chest) to collect 44,800 samples and train & deploy the neural network model using pure numpy. Accuracy: 97.5%. Code: [github]
- **Skills/Tools**: Python, C++, Raspberry-Pi, ESP8266. Dataset: [github].

Lempel-Ziv-Welch Text File Compression - A python package **Apr'18 - Sept'18**

- Designed and implemented a python package for UTF-8 file compression
- Achieved an average compression ratio of 50% along with O(log(n)) phrase look-up time using the Trie data structure. Link to repository: [github]
- Studied compression ratio as a function of file probability distribution by generating and compressing synthetic files with Exponential, Poisson, Uniform and Gaussian distributions.

Publications

- H. Kale, **P. Mandke**, H. Mahajan, V. Deshpande,"Human Posture Recognition using Artificial Neural Networks", 2018 IEEE 8th International Advance Computing Conference, Greater Noida, India, 2018, pp. 272-278.
  URL: https://ieeexplore.ieee.org/document/8692143

Skills

Primary: Python, PyTorch, Tensorflow, Numpy, Git, Linux, Docker
Secondary: C++, Kotlin, gRPC, ROS, MATLAB

Awards & Honors

- Recipient of the prestigious **Narotam Sekhsaria Scholarship** for graduate studies in 2019.
  Details: https://pg.nsfoundation.co.in/